

How can we detect cheating effectively without compromising system security through kernel-level access?

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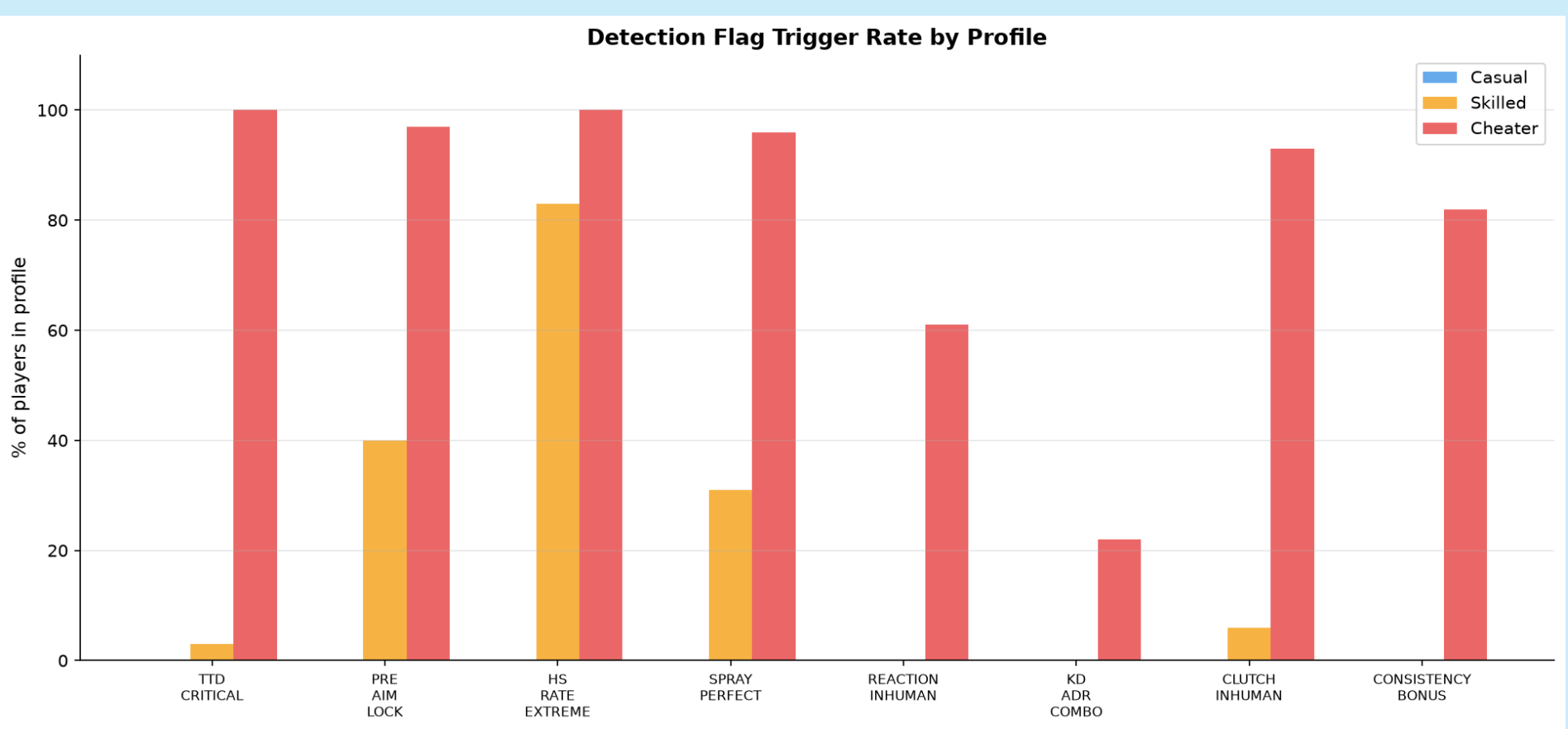
Problem

Kernel-level anti-cheats (Ring 0) are becoming the industry standard for stopping cheaters in competitive FPS games such as Counter-Strike 2, but they require granting a driver unrestricted access to the entire system, not just the game. This creates three risks:

- **Security:** expands the attack surface
- **Privacy:** violates GDPR principles of data minimization
- **Trust:** drives boycotts, excludes Linux users

This thesis explores an alternative: detecting cheaters through server-side behavioral analysis of gameplay telemetry, without ever touching the player’s machine.

Detection Flags



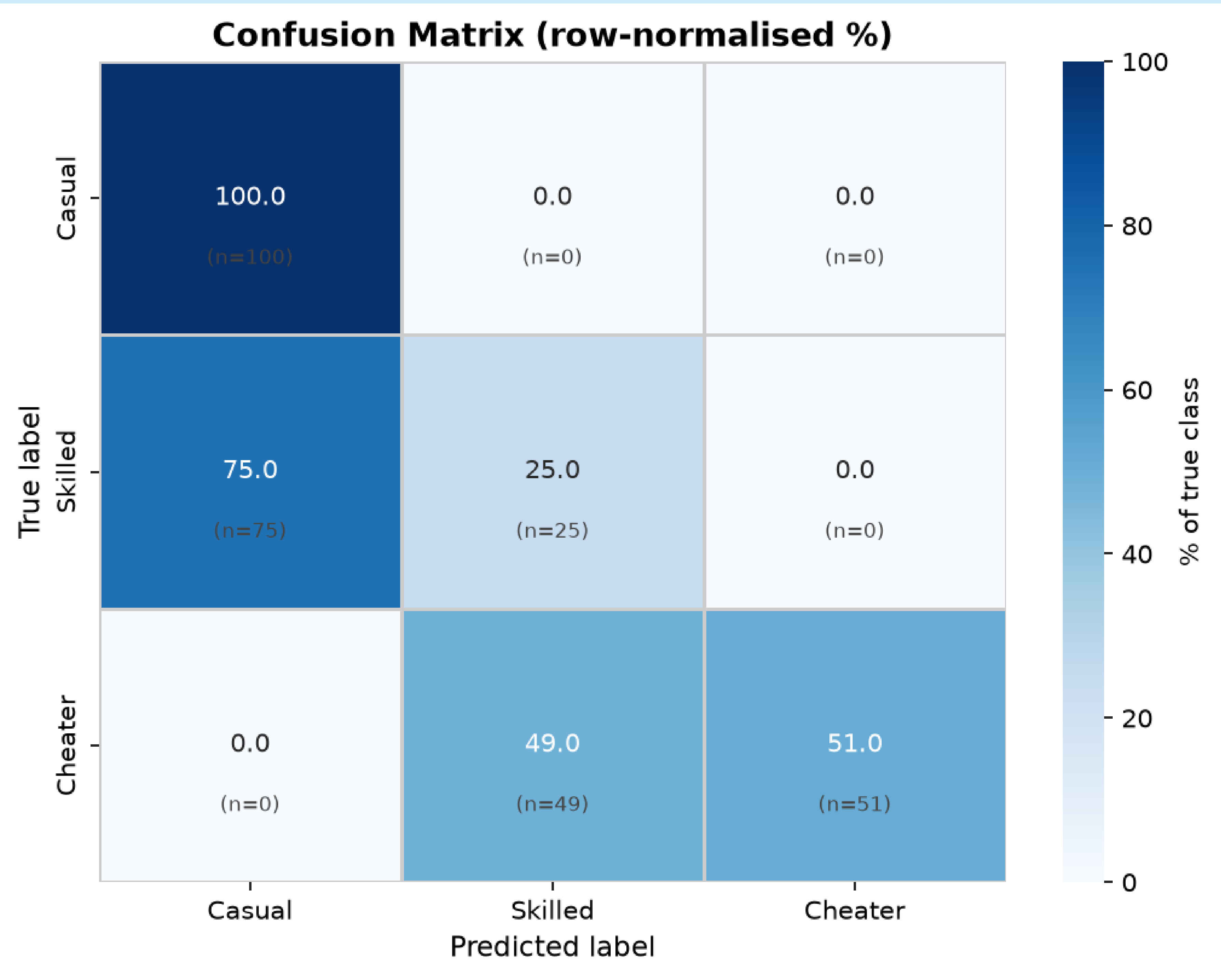
Five flags fire in over 90% of cheater sessions: four stay silent on skilled players, but HS_RATE also fires on 83% of skilled sessions, exposing it as the least significant signal on its own. Cross-session persistence across *several* flags is what separates cheaters from skilled players [LLB25].

Kernel-Level Risks

- Kernel-level anti-cheat is stuck in a structural arms race it cannot win: cheat vendors patch around detections in days, while publishers need weeks to safely ship a kernel update [KF24]. Worse, it is categorically blind to DMA cheats.
- To test how far this privileged access actually reaches, Windows-Kernel-Explorer and Cheat Engine were used on an isolated VM to locate and reconstruct a password from an *unrelated* Notepad process’s memory. Ring 0 access is not just theoretically broad: it is critically intrusive against any parallel process, like browsers or password managers.

Results (synthetic benchmark)

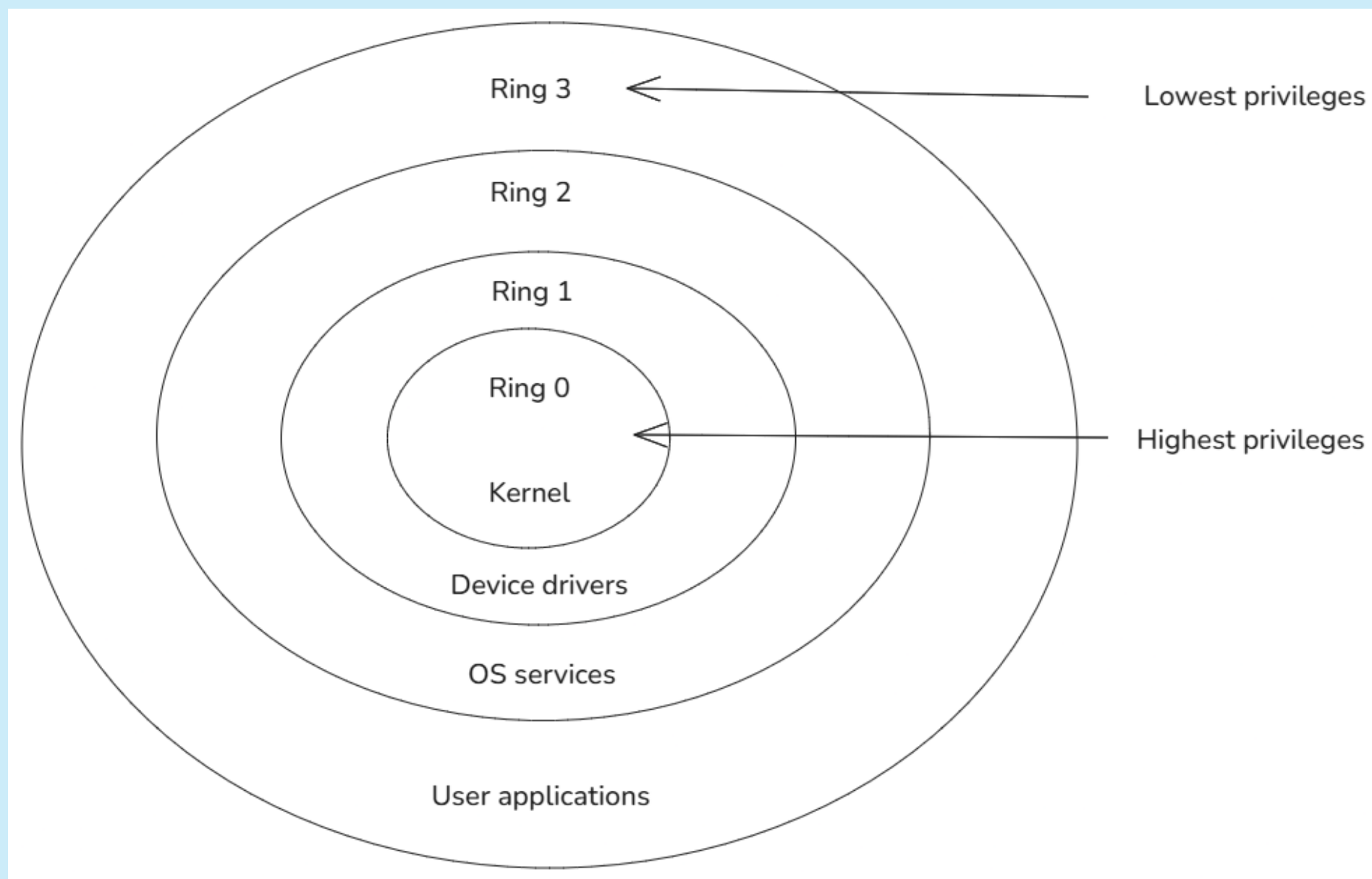
Zero false positives, 51% recall: a conservative threshold flags half of cheaters automatically, the rest rank just below it for review.



Conclusion:

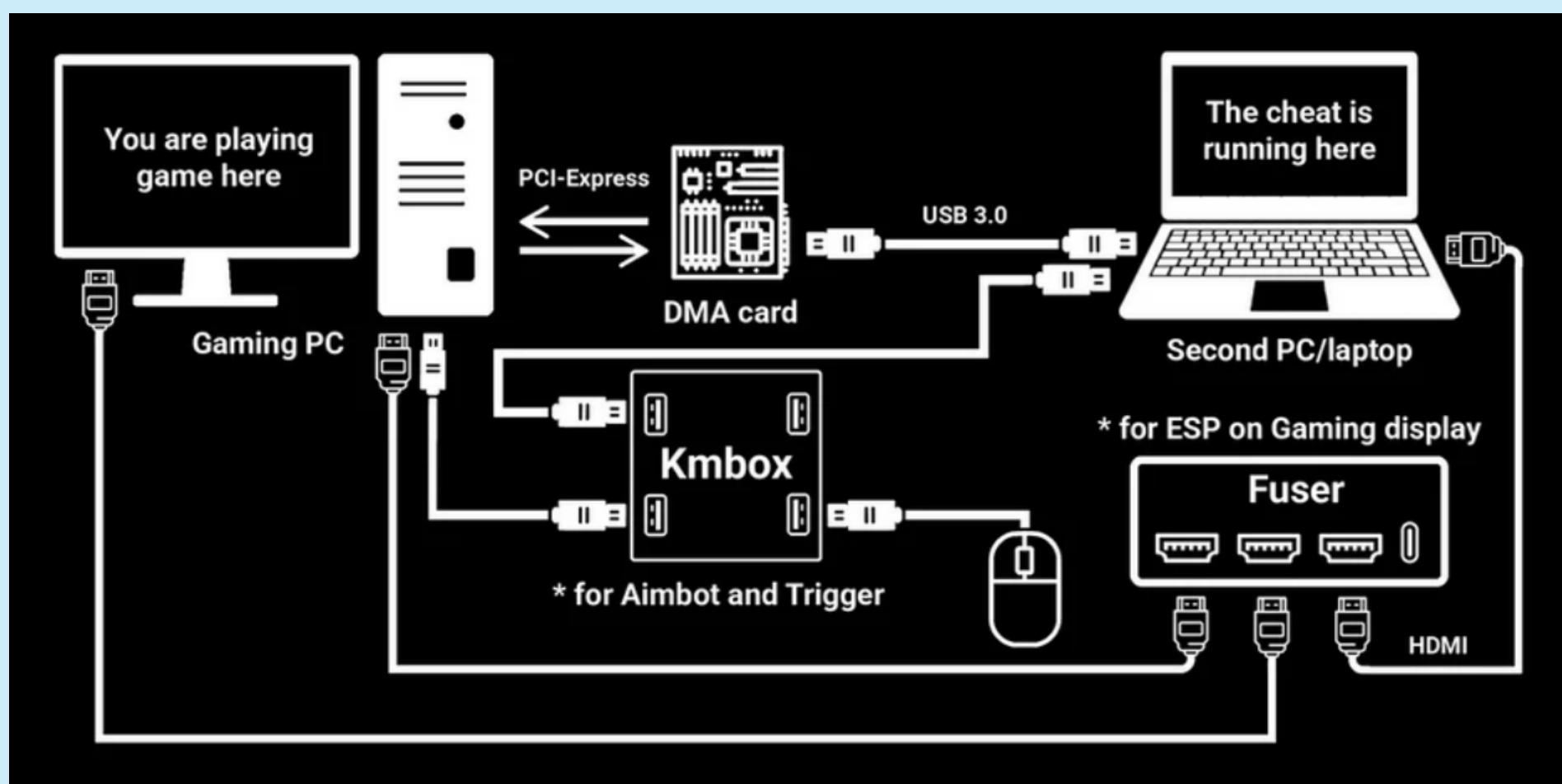
Kernel-level and server-side detection fail on opposite classes of cheat (one blind to hardware-level manipulation, the other blind to a cheat used once and never again), making them complementary layers rather than substitutes [DK24; Col+24].

Privilege Rings



Kernel-level anti-cheats and kernel-level cheats compete for the same Ring 0 authority, leaving no visibility to user-mode software.

The DMA Blind Spot



A PCIe card reads host RAM directly, a second PC computes and injects synthetic input. No driver loads, no process is touched: the kernel-level anti-cheat has nothing to detect.

Suspicion Score

Two independent layers combine into one suspicion score per session:

$$\text{score} = 0.6 * \text{heuristic} + 0.4 * (100 * \text{lstm})$$

heuristic: 8 threshold-calibrated flags (time-to-damage, pre-aim, reaction time, KD/ADR, clutch, consistency, etc).

lstm: a probability modeling round-by-round behavioral persistence. A flag only counts if it recurs across a player’s match history, not a single session.

References

[KF24] Y. Kuzhii and Y. Furgala. “THE ANTICHEAT DEVELOPMENT IN THE COUNTER-STRIKE MAIN SERIES”. In: *Electronics and Information Technologies*, vol. 26. 2024. URL: <https://doi.org/10.30970/e11.26.8>.

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[Col+24] Sam Collins et al. “Anti-Cheat: Attacks and the Effectiveness of Client-Side defenses”. In: *Proceedings of the 2024 Workshop on Research on offensive and defensive techniques in the context of Man At The End (MATE) attacks*. ACM, pp. 30–43, 2024. URL: <https://doi.org/10.1145/3689934.3690816>.

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